TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	81.3 ± 1.62
Correlation Stability	90.8 ± 0.87
Distribution Consistency	90.6 ± 1.11
Entropy Preservation	87.3 ± 1.49
Mean Z Score Drift	0.2 ± 0.04
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.3 ± 0.46
Data Density	100.0 ± 0.00
Coverage Score	81.4 ± 3.83
Feature Accuracy	82.3 ± 3.20
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Pregnancies	3.845	4.088	+0.243	6.3%	OK
Glucose	120.895	114.233	-6.662	5.5%	OK
BloodPressure	69.105	72.244	+3.139	4.5%	OK
SkinThickness	20.536	23.352	+2.815	13.7%	Warn
Insulin	79.799	41.707	-38.093	47.7%	Drift
BMI	31.993	35.140	+3.148	9.8%	OK
DiabetesPedigreeFunction	0.472	0.291	-0.180	38.2%	Drift
Age	33.241	30.218	-3.023	9.1%	OK
Outcome	0.349	0.316	-0.033	9.5%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	81.3 ± 1.62
Correlation Stability (*2)	90.8 ± 0.87
Distribution Consistency (*3)	90.6 ± 1.11
Entropy Preservation (*4)	87.3 ± 1.49
Mean Z Score Drift (*5)	0.2 ± 0.04
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.3 ± 0.46
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	81.4 ± 3.83
Feature Accuracy (*10)	82.3 ± 3.20

Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7493506493506492	0.7298701298701299	0.7538961038961042	0.00454545454545454	
F1	0.6076604413342895	0.6246625575820672	0.6265943050207405	0.01893386368645097	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7538961038961043	0.7402597402597403	0.7590909090909091	0.00519480519480508	
F1	0.6155511296147973	0.6355115231687594	0.6276394258908036	0.01208829627600638	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7461038961038961	0.7311688311688311	0.7480519480519481	0.00194805194805195 43	
F1	0.6159022356442937	0.6326368496760761	0.6193383272720822	0.00343609162778832 9	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6675324675324675	0.6571428571428571	0.7064935064935065	0.03896103896103895	
F1	0.4907509554720443	0.4934200713978905	0.5337854799993638	0.04303452452731950 5	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7720779220779222	0.7487012987012986	0.7597402597402597	-0.012337662337662337	
F1	0.6262914445612621	0.6449006708177021	0.6262455545225578	-4.589003870432151e-05	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7564935064935064	0.7441558441558442	0.7584415584415584	0.00194805194805192	
F1	0.5733680734057942	0.5840458958430255	0.6006711423077422	0.02730306890194803	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7292207792207792	0.7272727272727273	0.7259740259740260	-0.00324675324675326 43	
F1	0.5921520638892818	0.6302140500865998	0.5955048704613639	0.00335280657208295 6	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7636363636363636	0.6935064935064935	0.7409090909090909	-0.02272727272727274 2	*
F1	0.6399852354635025	0.6327553522969641	0.6406895029912094	0.00070426752770695 88	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7149350649350657	0.6727272727272727	0.6785714285714286	-0.03636363636363636	
F1	0.5748649566274736	0.5617859017319466	0.5471614521839192	-0.02770350444355366 3	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7545454545454545	0.7253246753246753	0.7525974025974026	-0.00194805194805194 3	
F1	0.5998259463389416	0.5488938411743838	0.6072663252368363	0.00744037889789466 15	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	98.0	66.0	34.0	90.0	43.54519302524994	0.27421814116927773	22.0	0.0	1.0
2	1.0	154.0	69.0	37.0	150.0	38.68533813462437	0.35217689578579023	25.0	0.0	1.0
3	0.0	96.0	60.0	17.0	143.0	26.060031612072184	0.31591004141121426	23.0	0.0	1.0
4	0.0	106.0	55.0	36.0	49.0	35.37461512363367	0.31157855212763924	21.0	0.0	1.0
5	10.0	163.0	67.0	1.0	8.0	47.098645048841 8	0.16085433311836722	33.0	1.0	1.0
6	5.0	69.0	57.0	0.0	-3.0	31.648406331999915	0.21733126596463934	35.0	0.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	93.0	74.0	33.0	74.0	28.145358901917 52	0.11653281624437561	23.0	0.0	2.0
2	1.0	102.0	84.0	34.0	164.0	30.842051689120346	0.30761841342939733	24.0	0.0	2.0
3	12.0	109.0	83.0	2.0	-1.0	29.651733914283398	0.27822430425515 47	52.0	1.0	2.0
4	4.0	82.0	76.0	-2.0	-3.0	27.838685893329888	0.21874503418013796	34.0	0.0	2.0
5	2.0	129.0	77.0	39.0	1.0	41.847319717562 19	0.28447652424126135	24.0	0.0	2.0
6	-1.0	147.0	63.0	38.0	192.0	34.260060950002334	0.55496889709651 55	25.0	1.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	112.0	70.0	33.0	106.0	38.358777914268984	0.33658224476763915	37.0	0.0	3.0
2	8.0	109.0	76.0	20.0	-5.0	38.502979807145804	0.2351297794627802	37.0	1.0	3.0
3	6.0	129.0	73.0	1.0	-3.0	27.463802989131608	0.3372873993251019	42.0	1.0	3.0
4	7.0	164.0	80.0	33.0	10.0	38.927029059942209	0.3105008484338664	42.0	1.0	3.0
5	2.0	79.0	68.0	33.0	71.0	35.744214574917161	0.4787419207362209	27.0	0.0	3.0
6	-1.0	88.0	68.0	32.0	-5.0	33.850629817860511	0.2096679791504424	40.0	0.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	92.0	89.0	25.0	59.0	28.399697091268537	0.22906876640363822	22.0	0.0	4.0
2	10.0	96.0	68.0	0.0	-2.0	31.714016914169223	0.2585049575507424	49.0	0.0	4.0
3	9.0	193.0	84.0	22.0	124.0	29.956705551712886	0.7255581096846521	46.0	1.0	4.0
4	2.0	96.0	53.0	3.0	1.0	36.602550978540202	0.3053460401682308 6	26.0	0.0	4.0
5	0.0	85.0	74.0	33.0	93.0	41.9498736526355	0.3212523420252881	25.0	0.0	4.0
6	2.0	90.0	77.0	-1.0	1.0	29.089689529742433	0.2731358873153733	19.0	0.0	4.0
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	84.0	84.0	30.0	128.0	36.2858459502046 9	0.22147162090703 168	21.0	0.0	5.0
2	2.0	97.0	61.0	44.0	154.0	29.0264730590777 55	0.07386209133894 736	25.0	0.0	5.0
3	1.0	97.0	63.0	22.0	43.0	24.9649542814520 66	0.19775475033962 7	26.0	0.0	5.0
4	1.0	85.0	65.0	33.0	10.0	21.6516146763766 56	0.17829177852100 433	24.0	0.0	5.0
5	7.0	98.0	82.0	0.0	-10.0	34.9550373185447 8	0.21257782377529 62	42.0	0.0	5.0
6	9.0	106.0	68.0	0.0	-5.0	27.5571048319180 8	0.26167140840151 7	44.0	0.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	103.0	74.0	28.0	169.0	29.4882486043898 95	0.25090156118308 47	27.0	0.0	6.0
2	1.0	111.0	69.0	31.0	-5.0	32.1466123594922 95	0.29761999149079 965	28.0	0.0	6.0
3	5.0	118.0	71.0	33.0	-1.0	23.2104684974865 77	0.17914351626659 758	23.0	0.0	6.0
4	3.0	127.0	88.0	33.0	142.0	38.2298611486587 1	0.70149975417834 62	29.0	1.0	6.0
5	6.0	99.0	50.0	-1.0	0.0	33.6977385852103	0.44154073864163 86	26.0	1.0	6.0
6	1.0	112.0	75.0	35.0	-4.0	34.0226325782577 3	0.58549183787727 01	25.0	1.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	-1.0	103.0	68.0	28.0	0.0	27.842911759359815	0.40453097493696855	20.0	0.0	7.0
2	3.0	139.0	68.0	1.0	-2.0	43.11887736970985	0.43318338646943794	25.0	1.0	7.0
3	8.0	102.0	85.0	-1.0	-9.0	27.379121493648704	0.32227607062230099	45.0	0.0	7.0
4	3.0	78.0	85.0	38.0	62.0	34.847080531161004	0.21499284649926395	25.0	0.0	7.0
5	3.0	182.0	63.0	0.0	0.0	33.046117520895	0.34038367726167884	39.0	1.0	7.0
6	0.0	94.0	72.0	43.0	157.0	24.667397230818644	0.62839400691131559	24.0	0.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	108.0	77.0	33.0	6.0	38.05480208192268	0.14875559293861035	27.0	0.0	8.0
2	1.0	98.0	85.0	41.0	-6.0	33.14994821756359	0.36776685672691041	24.0	0.0	8.0
3	1.0	99.0	75.0	47.0	139.0	38.80353470953022	0.50139495628654092	22.0	1.0	8.0
4	0.0	92.0	73.0	27.0	107.0	30.442530307045235	0.3464380776843688	21.0	0.0	8.0
5	2.0	85.0	71.0	32.0	139.0	46.648014088032901	0.23976244555731804	24.0	0.0	8.0
6	1.0	96.0	68.0	35.0	-2.0	39.340215531759099	0.21603806844512752	54.0	0.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	147.0	76.0	47.0	191.0	34.2928496233800 5	0.79342061472301 04	23.0	1.0	9.0
2	7.0	91.0	76.0	31.0	-4.0	30.2148607687166 2	0.32179680025009 394	43.0	0.0	9.0
3	4.0	109.0	77.0	25.0	-11.0	31.7728195531059 54	0.19800969677509 614	20.0	0.0	9.0
4	1.0	94.0	70.0	21.0	-6.0	30.9285270705088 95	0.13593538956427 192	23.0	0.0	9.0
5	6.0	102.0	63.0	-1.0	1.0	30.9856169531954 6	0.12445563050122 871	26.0	0.0	9.0
6	4.0	90.0	81.0	19.0	193.0	32.8053614854014 55	0.39065298443798 06	31.0	0.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	11.0	105.0	78.0	30.0	-7.0	37.8026087518588	0.28200611631332 08	32.0	1.0	10.0
2	5.0	106.0	78.0	1.0	-4.0	31.8777950805137 96	0.26594993655724 347	50.0	1.0	10.0
3	1.0	124.0	73.0	19.0	134.0	33.1929346475565 8	0.64131894297430 98	29.0	0.0	10.0
4	1.0	127.0	63.0	34.0	-3.0	32.7157753604687 7	0.39279732741450 77	52.0	1.0	10.0
5	9.0	99.0	78.0	29.0	0.0	36.6366746939292 4	0.21160128278462 784	63.0	1.0	10.0
6	6.0	114.0	74.0	1.0	1.0	31.9274065170806 27	0.14950263987830 98	23.0	0.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan